Strongly connected orientations

A mixed graph is a graph in which some edges may be directed and others undirected. There is at most one edge between every pair of vertices, which may be directed or undirected. An orientation of a mixed graph is a directed graph obtained by assigning exactly one of the two possible directions to an undirected edge. Given a mixed graph, you have to determine whether it has an orientation that is strongly connected, and if so, find the directions to be assigned to each undirected edge to get such an orientation. A directed graph is strongly connected if there is a directed path from any vertex to any other.

Input/ Output Format
The first line of input contains two integers n (number of vertices) and m (number of edges) separated by a space. Each of the next m lines contains three integers i, j, t separated by a space, indicating an edge between vertices i and j, where $0 \leq i, j < n$ and $i \neq j$. If $t = 0$ the edge is undirected and if $t = 1$, it is directed from i to j. It is given that $n \leq 10^6$ and $m \leq 10^7$.

The first line of output should be "possible" if there exists such an orientation, otherwise "impossible" (without quotes). If possible, the subsequent lines should give the directions to be assigned to each undirected edge to make the graph strongly connected. Each line should print the endpoints of one undirected edge such that the direction assigned is from the first vertex to the second. Any valid solution can be output. You will get half credit if you can decide correctly whether it is possible or not.

Sample Input/Output
Input 1
4 4
0 1 0
1 2 0
2 3 0
0 3 1
Output 1
possible
1 0
2 1
3 2
Input 2
4 4
0 1 1
1 2 0
2 3 0
0 3 1
Output 2
impossible